

Dashboard Summary

1. Dashboard Purpose

The dashboard is designed as a self-serve, interactive analytics platform that allows stakeholders to independently evaluate regional crime distribution, temporal trends, and socio-economic drivers across the following four police forces: Metropolitan, Surrey, West Midlands, Northumbria.

- **Multi-Dimensional Filtering:** Provides seamless interactive slicing across Police Force Areas, Local Authority Districts (LADs), crime categories, years, and specific months.
- **Core Analytics:** Evaluates overall crime volumes, population-normalized rates (per 1,000 residents), and localized growth trajectories across a continuous 3-year observation window (June 2023 – May 2026).
- **Socio-Economic Deep Dive:** Houses a dedicated analytical dashboard investigating the direct relationship between Index of Multiple Deprivation (IMD) deciles and crime concentration at the LSOA level.
- **Operational Hotspot Tracking:** Identifies specific local authorities demonstrating positive upward growth trends (slopes) to support proactive police resource deployment.

2. Intended Audience

The platform is structured to serve a dual-audience profile, balancing visual clarity for operational officers with rigorous statistical depth for strategic data analysts.

- **Primary User : Operational Police Officers & Command Staff:** Tailored for time-poor decision-makers who require rapid, at-a-glance situational awareness. High-level KPI cards, intuitive spatial map grids, and clean visual hierarchies deliver immediate key takeaways without requiring complex data manipulation. The first two dashboards of my analysis are most suited for non data literate users.
- **Secondary User : Data Analysts & Policy Strategists:** Regression analysis particularly in my third dashboard requires slightly more technical understanding, while it was explained in my presentation and I do believe it is simple enough to be worked by non-technical individuals, I would be open to any questions about how it works and to walk people through its use.

3. Key Assumptions & Limitations

To ensure proper interpretation of insights and avoid over weighing the discovered insights, stakeholders should consider the following core methodological assumptions and data limitations:

- **Police-Recorded Data Reliance:** The underlying dataset captures exclusively reported, police-recorded incidents. Unreported offences ('dark figure' of crime) and historical variations in force-specific recording practices are not reflected.

- **Static Population Benchmarking:** Crime rates per 1,000 residents rely on static ONS residential population baseline data. This metric does not account for transient populations (e.g., daily commuters, tourists, and nighttime economy patrons), which artificially inflates per-capita rates in commercial and entertainment hubs such as Westminster.
- **Cross-Sectional Regression Design:** The IMD decile regression measures spatial cross-sectional correlation across neighborhoods rather than utilizing a dynamic panel time-series regression model, limiting the ability to establish temporal causality. The analysis described would be ideal, but that data is very hard to come by and is outside the scope of my analysis.
- **Sample Window Scope:** Analysis is bounded by a 36-month observation window (June 2023 – May 2026). While highly effective for evaluating current operational trends, it may not reflect structural shifts.
- **Correlation vs. Causation:** While IMD deciles account for approximately 45% of variance in log violent crime rates ($R^2 \approx 0.453$), it is important to note that this analysis is slightly circular. In fact, 11% of the weighting of an IMD score is precisely crime. Therefore, we would expect to see higher values of R^2 and they should be taken with a pinch of salt. Moreover, this analysis captures a static three year period of time, where IMD has remained constant due to restrictions in the data, to conduct a proper analysis on the link between IMD deprivation scores and crime rate I would recommend a timeseries regression analysis using specialised models such as ARMA.
- There is also reason to debate the validity of using IMD as a metric for regression analysis at all, in particular, IMD is computed by ranking all LSOA districts and then splitting them into deciles, where 1 is the most deprived and 10 is the least deprived. This means that we have no 'objective' sense of depravity because all LSOAs are compared to each other. In practice, perhaps we can conduct a regression analysis using an instrumental variable to avoid dealing with endogeneity and IMD limitations. Again, this is far outside the scope of my analysis and would probably be worthy of a research paper, my regression analysis was simply designed to highlight and illustrate a potential issue and surface level relationship to prompt further research. It is not intended as a rigorous score.